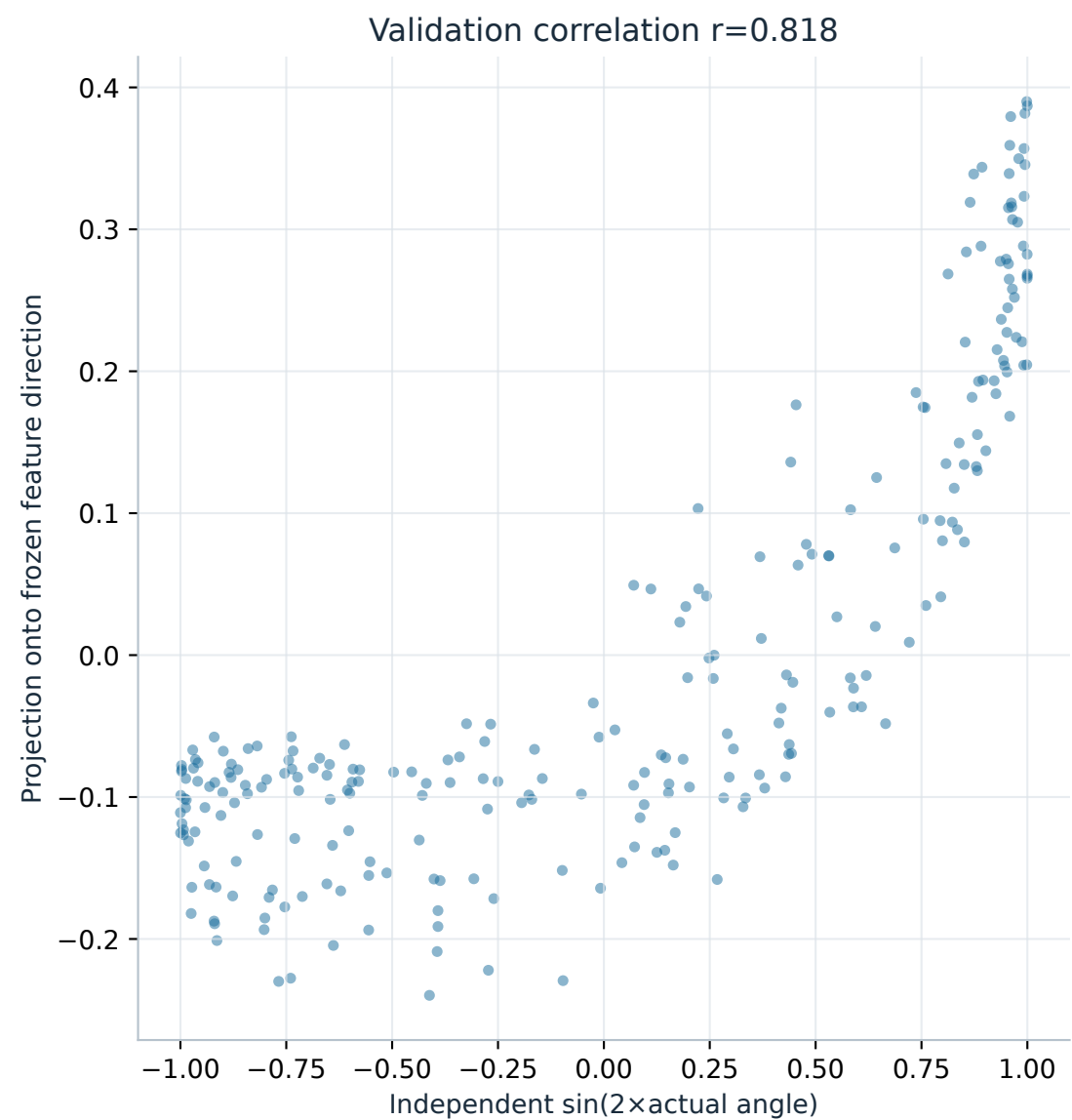


What the feature pulse does—and does not—identify



Calibration and causal limits

- 64 fit trials; 64 independent validation trials
- Activation RMS: 0.4574
- 32-channel direction has RMS1
- Absolute orientation preference, not change preference
- No main-model gradients, optimizer or training
- Native final output head only
- One checkpoint; no population-level biological claim

OLS encoding localizer uses actual $\cos(2\theta)/\sin(2\theta)$ orientation on independent native sample frames; trial split precedes pooling locations. Cue location/sign and task label are not calibration targets. Absolute sample orientation is independently drawn from the task label/cue rule. Positive validation confirms feature preference, not a signed rotation comparator or universal behavioral enhancement.